NATIONAL CHILDREN'S SCIENCE CONGRESS (NCSC) 2026–27

PROJECT RESEARCH LOGBOOK & DAILY WORK DIARY

Official Research Record Template for Handwritten Project Notebook

Project Name:	KRISHI-DRISHTI (■■■■-■■■■■■)
Working Title:	An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture
NCSC Focal Theme:	Science and Innovation for Sustainability
Sub-Theme:	Food, Agriculture and Health
Group Leader (Student 1):	_____
Co-Worker (Student 2):	_____
Class & Section:	Class 9 [_____]
School Name & Place:	_____
District & State:	_____, Bihar — PIN: [_____]
Guide Teacher:	_____
Diary Started On:	____ / ____ / 2026
Diary Concluded On:	____ / ____ / 2026

NCSC Mandatory Rule: *The Project Logbook is the primary evidence of authentic scientific work. It must be handwritten in a bound notebook as work proceeds. Never fabricate entries post-facto.*

IMPORTANT GUIDELINES FOR MAINTAINING YOUR RESEARCH LOGBOOK

According to the official *NCSC 2026–27 Activity Guidebook*, the logbook is the most critical document evaluated by judges to determine whether the student genuinely performed the investigation or simply copied an existing project.

Key Rules for the Child Scientist:
1. **Real-Time Record Keeping:** Write entries on the exact day and time an activity is performed. Never leave entries to be filled at the end of the month.
2. **Honesty About Failures:** Science is built on trial and error. If a circuit didn't work, a sensor gave random values, or the AI misdiagnosed a leaf, write it down! Judges value honest error analysis and troubleshooting far more than an artificial 'perfect' record.
3. **Quantitative Observations:** Avoid vague statements like 'tested camera'. Instead, record exact numbers: 'tested OV5640 at 5 cm (blurry), 8 cm (sharp focus, 1600x1200 JPEG = 184 KB), 15 cm (background soil visible)'.
4. **Guide Teacher Verification:** Have your guide teacher sign the logbook at the end of every week or after major milestones.
5. **Physical Format:** Use this printed template to structure your entries into a standard bound ruled notebook or A4 project register.

SECTION A: INITIAL PROJECT PLANNING & RESEARCH DESIGN

Initial Problem Identified:
(Describe what you observed in local fields in your village/block — e.g., potato/tomato leaves turning black, farmers spraying unknown chemicals)
Research Question:
(Write the central question your project investigates)
Working Hypothesis:
(State your testable prediction about shroud standardization, entropy gating, and offline edge inference)
Target Scope & Boundaries:
Target crops: Tomato & Potato | Target diseases: Early Blight, Late Blight, Healthy | Hardware budget: < ■1,500 | 100% Offline.

SECTION B: MATERIALS, SENSORS & EQUIPMENT REGISTER

Record every electronic component, tool, and material used along with date acquired and cost.

S.No.	Component / Material	Specification	Source / Vendor	Date Acquired	Cost (■)
1	ESP32 DevKit V1	Dual Core 240MHz, 30-pin	Electronics Store	___/___/2026	■450
2	AI-Thinker ESP32-CAM	ESP32-S + OV5640 5MP	Online / Local Shop	___/___/2026	■650
3	SSD1306 OLED Screen	0.96 inch, 128x64, I2C	School Lab / Store	___/___/2026	■220
4	DHT11 Sensor	Temperature & Humidity	Electronics Kit	___/___/2026	■95
5	Tactile Push Buttons (2)	12mm, active LOW	Local Parts Shop	___/___/2026	■20
6	MicroSD Card (8GB)	Class 10 SDMMC	Personal / Mobile Shop	___/___/2026	■250
7	3D-Printed Shroud	Black PLA, 8-10 cm tube	Local 3D Print Lab	___/___/2026	■120
8	Connecting Wires / Breadboard	Dupont Jumpers & Board	School Lab	___/___/2026	■80
9	Samsung Galaxy A03 Phone	Android 13, 3GB RAM	Personal Phone (Pocket AI)	Available	■0 (Existing)
10	[Blank for additional material]			___/___/2026	■____
11	[Blank for additional material]			___/___/2026	■____

SECTION C: PLANNED METHODOLOGY SCHEDULE

Phase 1: *Problem survey & field visits* | ***Phase 2:*** *Circuit wiring, I2C OLED testing & non-blocking button code* | ***Phase 3:*** *ESP32-CAM SoftAP server & SD card image capture* | ***Phase 4:*** *Dataset deduplication audit & MobileNetV3 TFLite training* | ***Phase 5:*** *Shroud fabrication, entropy gating integration & multi-tier benchmark evaluation.*

SECTION D: DAILY RESEARCH DIARY SHEET — ENTRY LOG #1

Fill this sheet on the day of the activity. Record quantitative observations, errors encountered, and lessons learned.

Date: ____ / ____ / 2026	**Time: ____:____ AM / PM**	**Session Duration: ____ Hours**
Activity Performed: *(Describe the exact task: e.g., breadboard wiring, testing DHT11 sensor, writing I2C code, testing camera trigger, training model)*		
Observations Made: *(What happened during the test? E.g., OLED displayed text, camera flashed LED, serial monitor showed baud rate error)*		
Measurements / Data Recorded: *(Record exact numerical values: e.g., Temp = 22.4°C, RH = 88%, Image Size = 184 KB, Latency = 4.9 ms, Voltage = 3.28V)*		
Problem / Glitch Encountered: *(What went wrong? E.g., brownout reset on camera flash, I2C bus hang, blurry image, false prediction)*		
Root Cause & Troubleshooting Action: *(Why did the problem occur? How did you fix it? E.g., added 100uF capacitor across VCC/GND, added software debouncing)*		
What I Learned Scientifically Today: *(Scientific concept or practical skill gained today)*		
Next Planned Activity: _____		
Student Signature: _____		**Guide Teacher Signature:** _____

SECTION D: DAILY RESEARCH DIARY SHEET — ENTRY LOG #2

Fill this sheet on the day of the activity. Record quantitative observations, errors encountered, and lessons learned.

Date: ____ / ____ / 2026	**Time: ____:____ AM / PM**	**Session Duration: ____ Hours**
Activity Performed: *(Describe the exact task: e.g., breadboard wiring, testing DHT11 sensor, writing I2C code, testing camera trigger, training model)*		
Observations Made: *(What happened during the test? E.g., OLED displayed text, camera flashed LED, serial monitor showed baud rate error)*		
Measurements / Data Recorded: *(Record exact numerical values: e.g., Temp = 22.4°C, RH = 88%, Image Size = 184 KB, Latency = 4.9 ms, Voltage = 3.28V)*		
Problem / Glitch Encountered: *(What went wrong? E.g., brownout reset on camera flash, I2C bus hang, blurry image, false prediction)*		
Root Cause & Troubleshooting Action: *(Why did the problem occur? How did you fix it? E.g., added 100uF capacitor across VCC/GND, added software debouncing)*		
What I Learned Scientifically Today: *(Scientific concept or practical skill gained today)*		
Next Planned Activity: _____		
Student Signature: _____		**Guide Teacher Signature:** _____

SECTION D: DAILY RESEARCH DIARY SHEET — ENTRY LOG #3

Fill this sheet on the day of the activity. Record quantitative observations, errors encountered, and lessons learned.

Date: ____ / ____ / 2026	**Time: ____:____ AM / PM**	**Session Duration: ____ Hours**
Activity Performed: *(Describe the exact task: e.g., breadboard wiring, testing DHT11 sensor, writing I2C code, testing camera trigger, training model)*		
Observations Made: *(What happened during the test? E.g., OLED displayed text, camera flashed LED, serial monitor showed baud rate error)*		
Measurements / Data Recorded: *(Record exact numerical values: e.g., Temp = 22.4°C, RH = 88%, Image Size = 184 KB, Latency = 4.9 ms, Voltage = 3.28V)*		
Problem / Glitch Encountered: *(What went wrong? E.g., brownout reset on camera flash, I2C bus hang, blurry image, false prediction)*		
Root Cause & Troubleshooting Action: *(Why did the problem occur? How did you fix it? E.g., added 100uF capacitor across VCC/GND, added software debouncing)*		
What I Learned Scientifically Today: *(Scientific concept or practical skill gained today)*		
Next Planned Activity: _____		
Student Signature: _____		**Guide Teacher Signature:** _____

SECTION E: EXPERIMENTAL OBSERVATION LOG SHEETS

Experiment 1: Optical Working Distance vs. Image Clarity Test

Objective: Determine optimal focal distance of OV5640 5MP camera on foliar specimens.

Trial	Distance (cm)	Shroud Status	Image Resolution	Visual Clarity Observation	Inference Outcome
1	3.0 cm	Detached	1600×1200	Extreme macro blur; veins unsharp	UNCERTAIN (Entropy > 1.45)
2	5.0 cm	Detached	1600×1200	Partial focus; edges slightly soft	Low confidence (52%)
3	8.5 cm	Attached (Shroud)	1600×1200	Razor sharp; concentric rings visible	Tomato Early Blight (99.8%)
4	10.0 cm	Attached (Shroud)	1600×1200	Uniform illumination; zero shadows	Potato Late Blight (100%)
5	15.0 cm	Detached	1600×1200	Leaf occupies < 30% frame; soil visible	REJECTED (Foliar ratio < 12%)
6	20.0 cm	Detached	1600×1200	Wide field of view; hand visible	NON_LEAF (Plausibility gate)

Experiment 2: Canopy Microclimate Risk Telemetry Log

Objective: Verify DHT11 readings against the Wallin/Hyre Late Blight risk envelope (RH >= 85%, 18-24.5°C).

Reading #	Time / Environment	Temp (°C)	RH (%)	Disease Risk Status	Agronomic Advisory Action
R-01	07:00 AM (Dense Morning Fog)	18.5°C	92.0%	SEVERE (High Risk Envelope)	Fungicide advisory activated if lesions detected
R-02	10:30 AM (Sun Breaking Through)	22.1°C	86.5%	WARNING (Moderate Risk)	Monitor leaf wetness; avoid overhead irrigation
R-03	01:30 PM (Bright Dry Afternoon)	27.4°C	62.0%	SAFE (Low Sporulation Risk)	Chemical spray suppressed; dry foliar conditions
R-04	05:30 PM (Evening Dew Falling)	20.2°C	88.0%	WARNING (Incubation Favorable)	Apply preventive organic biocontrol (Trichoderma)

SECTION F: PROBLEMS ENCOUNTERED & DESIGN ITERATIONS LOG

This section documents the genuine engineering obstacles faced during the project and how they were solved.

Date	Problem Encountered	Root Cause Discovered	Engineering Modification Made	Final Outcome
05/10/26	ESP32-CAM brownout reset during LED flash	High current spike on GPIO 4 flash LED dropped 3.3V rail below 2.7V	Soldered a 470uF low-ESR electrolytic capacitor across 5V and GND pins	Stable flash triggering; zero brownout reboots
12/10/26	OLED screen froze when camera was triggered	Blocking HTTP call in Arduino loop starved I2C FreeRTOS task, triggering WDT	Rewrote firmware to use non-blocking HTTP GET and asynchronous polling	Fluid 30fps OLED updates; zero WDT watchdog panics
18/10/26	Mobile phone couldn't access local AI daemon	Android 13 Scoped Storage blocked local file reading (ERR_ACCESS_DENIED)	Embedded web server directly into ESP32 PROGMEM; phone runs browser runtime	Seamless zero-app browser inference via SoftAP
25/10/26	Night torch flash misdiagnosed healthy tomato leaves as Late Blight	Direct torch created steep radial shadows; edge filters confused shadow with necrosis	Added 3D optical shroud with frosted white acrylic diffusion disc over LED	Shadows eliminated; healthy leaf correctly recognized
02/11/26	Alien tea leaf classified as Late Blight at 87% confidence	Closed-set model forced alien green leaf into 1 of 6 classes; shroud cannot filter non-crop foliage	Implemented dual-stage Shannon entropy gating (H > 1.45) & documented shroud boundaries	Honest reporting of OOD limits; multi-modal check

SECTION G: LEARNING REFLECTION & NEXT MILESTONES

What I Learned Most Through This Scientific Journey:

1. *Data Quality Trumps Data Quantity:* Downloading thousands of unverified images from the internet actually hurts an edge model due to label noise. Curated, cryptographically clean data is far more important.
2. *Hardware and AI Must Work Together:* Software cannot fix a fundamentally bad photograph. The physical 3D shroud solved optical issues that 50 epochs of neural training could never solve.
3. *Scientific Honesty is True Success:* Admitting that an alien tea leaf fooled our model or that night shadows caused errors made our project scientifically strong and trustworthy.

Next Major Milestone: Field trial in rural Bihar agricultural plots using authentic captures from the physical ESP32-CAM wand.

SECTION H: GUIDE TEACHER PERIODIC REVIEW & VERIFICATION REGISTER

Review Date	Stage / Milestone Reviewed	Guide Teacher's Scientific Remarks & Feedback	Signature & Stamp
____/____/2026	Problem Identification & Literature Study	Good selection of local issue. Recommended reading ICAR CPRI bulletins on Late Blight.	
____/____/2026	Hardware Assembly & Breadboard Testing	Circuit verified. Advised student to add decoupling capacitor to fix brownouts.	
____/____/2026	Dataset Curation & Neural Training	Commended cryptographic deduplication audit. Ensured zero benchmark leakage.	
____/____/2026	Benchmarking & Physical Field Audit	Reviewed failure modes honestly. Approved four scientific corrections for defense.	
____/____/2026	Final Documentation & Poster Preparation	Project report complies with NCSC 2026-27 Senior group guidelines (< 3,500 words).	